Solid State Physics Practice

Budapest University of Technology and Economics

Problem Set 2 — Elastic Scattering

Fall 2026

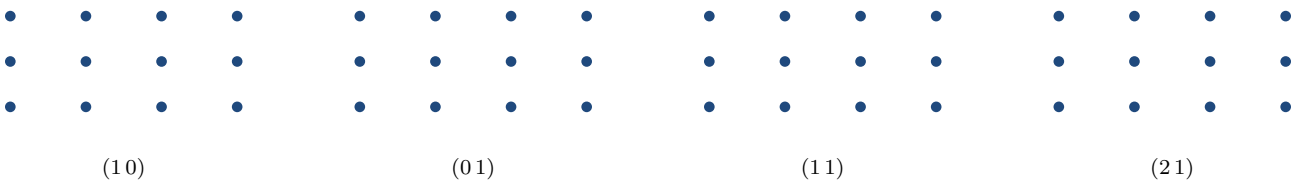
Keywords: Miller indices, lattice lines and interplanar spacing, elastic X-ray scattering, Laue condition, Bragg condition, reciprocal lattice, atomic form factor, structure factor, basis, extinction (systematic absences), powder diffraction, honeycomb structure (graphene, hBN), fcc and bcc lattices, CuO_2 plane, triangular lattice

Notation: $[uvw]$ denotes a real-space lattice vector or direction; (hkl) denotes a lattice plane or a reciprocal lattice vector. $\mathbf{G}_{hkl} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$ is a reciprocal lattice vector; $f(\Delta\mathbf{k}) = \int dV\, \rho(\mathbf{r})\, e^{i\Delta\mathbf{k}\mathbf{r}}$ is the atomic form factor; $S(\mathbf{G}) = \sum_\alpha f_\alpha e^{i\mathbf{G}\tau_\alpha}$ is the structure factor, $I \propto |S|^2$. The scattering angle is 2ϑ, so $\sin^2\vartheta = \lambda^2|\mathbf{G}|^2/16\pi^2$.

Exercise 1 — Miller Indices in a Two-Dimensional Lattice

Keywords: Miller indices, lattice lines, interplanar spacing, reciprocal lattice, rectangular lattice

Consider a two-dimensional rectangular lattice with primitive vectors $\mathbf{a}_1 = a\,\hat{\mathbf{x}}$ and $\mathbf{a}_2 = b\,\hat{\mathbf{y}}$ ($a \neq b$). A family of parallel lattice lines – the two-dimensional analogue of lattice planes – is labeled by a pair of integers $(h\,k)$, defined so that the lines intercept the two axes at a/h and b/k (the intercept is at infinity if the corresponding index is 0). Consecutive lines of a family are parallel and equally spaced by a distance d_{hk}. A patch of the lattice is shown four times below, once for each family you will need.



(a) On each patch of lattice points above, sketch the two lines of the labeled family $(h\,k)$ that lie closest to the origin, and mark which lattice rows they run along. Check your lines against the required intercepts a/h and b/k.

(b) Show that the reciprocal lattice vector $\mathbf{G}_{hk} = h\mathbf{b}_1 + k\mathbf{b}_2 = 2\pi\left(\frac{h}{a}, \frac{k}{b}\right)$ is perpendicular to the $(h\,k)$ family of lines, and that the spacing of the family is $d_{hk} = 2\pi/|\mathbf{G}_{hk}|$.

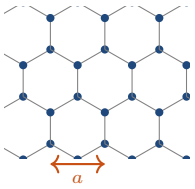
(c) Derive the general formula $\dfrac{1}{d_{hk}^2} = \dfrac{h^2}{a^2} + \dfrac{k^2}{b^2}$ for the interline spacing of the rectangular lattice.

(d) For $a = 3.2\,\text{Å}$ and $b = 4.5\,\text{Å}$, compute d_{10}, d_{01}, d_{11} and d_{21}. Which family has the largest spacing, and how does this relate to the density of lattice points lying on each line?

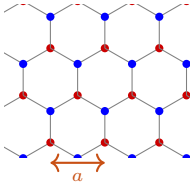
Exercise 2 — Graphene and Hexagonal Boron Nitride

Keywords: triangular lattice, two-atom basis, honeycomb structure, structure factor, extinction, intensity ratio

Graphene atoms are arranged in a honeycomb structure with lattice constant a, the distance between neighbouring equivalent atoms (see the left figure; the nearest-neighbour distance is $a/\sqrt{3}$): every atom sits at the vertex of a hexagon, with three nearest neighbors. This honeycomb net is *not* itself a lattice, since a translation that maps one atom onto its neighbor does not map the whole structure onto itself. Instead it is a two-dimensional *triangular* lattice decorated with a two-atom basis. Hexagonal boron nitride (hBN, right figure) has the same triangular lattice and the same honeycomb arrangement of sites, but one of the two basis atoms is boron and the other nitrogen.



Graphene (C, C)



hBN (B, N)

(a) Identify the triangular lattice underlying the honeycomb structure, choose primitive vectors, and give the two-atom basis that places atoms at every vertex of the hexagons. Find the reciprocal lattice vectors and $|\mathbf{G}_{hk}|$.

(b) Compute the structure factor of graphene (two identical atoms, form factor f). Is there any extinction? What is the intensity ratio of the peaks?

(c) Compute the structure factor of hBN with form factors $f_A \neq f_B$. When could an extinction occur? What is the intensity ratio of the weak and strong peaks if $f \propto Z$ ($Z_\mathrm{B} = 5$, $Z_\mathrm{N} = 7$)?

(d) Show that for $N = h^2 - hk + k^2$ the intensity depends only on N modulo 3. List the first few scattering rings of a powder sample with their relative intensities.

Exercise 3 — Powder Pattern of Cubic and fcc Crystals

Keywords: powder diffraction, simple cubic, fcc, primitive and conventional cells, reciprocal lattice vectors, systematic absences, lattice constant

A powder is made from a cubic crystal of lattice constant a, and an elastic scattering experiment is performed with a monochromatic X-ray beam of wavelength λ.

(a) A simple cubic lattice has primitive vectors $\mathbf{a}_1 = a\hat{\mathbf{x}}$, $\mathbf{a}_2 = a\hat{\mathbf{y}}$, $\mathbf{a}_3 = a\hat{\mathbf{z}}$. Derive the primitive reciprocal lattice vectors and show that $|\mathbf{G}_{hkl}| = \frac{2\pi}{a}\sqrt{h^2+k^2+l^2}$, so that a Bragg peak occurs only at angles satisfying

$$\sin^2\vartheta = \frac{\lambda^2}{4a^2}\left(h^2+k^2+l^2\right) = \frac{\lambda^2}{4a^2}N.$$

As h, k, l range over the integers, which values can N take? Find all the missing values of N up to $N = 16$.

(b) A face-centered cubic (fcc) lattice can be described by the primitive vectors

$$\mathbf{a}_1 = \tfrac{a}{2}[0,1,1], \qquad \mathbf{a}_2 = \tfrac{a}{2}[1,0,1], \qquad \mathbf{a}_3 = \tfrac{a}{2}[1,1,0].$$

Derive the primitive reciprocal lattice vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$, and show that $\mathbf{G}_{hkl} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$ satisfies

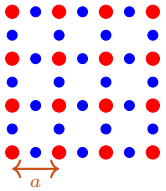
$$|\mathbf{G}_{hkl}|^2 = \frac{4\pi^2}{a^2}\left[3(h^2+k^2+l^2) - 2(hk+kl+lh)\right], \qquad \sin^2\vartheta = \frac{\lambda^2}{4a^2}N.$$

Which values of N occur for arbitrary integer triples (h,k,l)? Find the missing values up to $N = 12$. Then describe the same fcc lattice using the conventional cubic cell with a four-atom basis, compute the structure factor, and show that its extinctions reproduce exactly this sequence of allowed N.

(c) The first peaks of an unknown cubic powder appear at $2\vartheta = 38.5°,\ 44.7°,\ 65.1°,\ 78.3°$ ($\lambda = 1.54\,\text{Å}$). Is the lattice sc, bcc or fcc, and what is a?

Exercise 4 — Scattering from the CuO_2 Plane *Keywords: cuprates, square lattice with basis, spherical shell form factor, structure factor, intensity maxima and minima*

The CuO_2 plane is a square lattice of Cu atoms (red, surface charge density $\sigma_1 > 0$), with O atoms (blue, $\sigma_2 < 0$) at the midpoints of each Cu–Cu bond; the Cu–Cu distance is a. Model both kinds of atoms as spherical shells of radius r_0 with uniform surface charge.



(a) Model the two kinds of atoms as spherical shells of radius r_0, with uniform surface charge densities $\sigma_1 > 0$ and $\sigma_2 < 0$, respectively. Give the atomic form factor of the two atom types.

(b) Determine the unit cell and the reciprocal lattice.

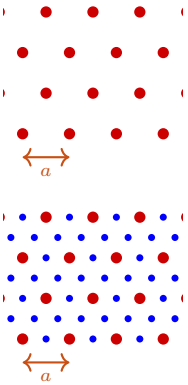
(c) Compute the structure factor characterizing the unit cell.

(d) For which scattering vectors $\Delta\mathbf{k}$ do we get intensity maxima and minima?

(e) What is the atomic form factor if the atoms are treated as spheres of radius r_0 with charge density $\rho(r) = r/(\pi r_0^4)$?

Exercise 5 — Triangular Lattice with One- and Multi-Atom Basis *Keywords: triangular lattice, layered crystal, powder diffraction, multi-atom basis, extinction*

In a crystal the atoms form triangular lattices of lattice constant a in planes perpendicular to the z axis (figure, top). The planes repeat along z at a distance $\frac{\sqrt{3}}{2}a$, with the atoms directly above each other. A powder is made from the crystal and an X-ray scattering experiment is performed.



(a) Using the primitive lattice vectors, determine the scattering angles ϑ at which finite scattered intensity is obtained, based on the Bragg condition.

(b) Suppose the basis of each triangular lattice becomes multi-atomic, as shown in the figure below: B atoms (red) at the lattice points, A atoms (blue) at the midpoints between neighbouring B atoms. Give the value of the structure factor at the reciprocal lattice vectors. For which ratio of the atomic form factors f_A and f_B, and at which reciprocal lattice vectors, is there extinction due to the structure factor?

(c) At which scattering angles ϑ does the scattered intensity vanish because of the structure factor?